

Volume 15 Runtime Ecosystem - Runtime Ecosystem Part 2

Version v77 draft

README.md

Runtime Ecosystem Part 2

Version: v77 draft

Status: Draft

Document ID: MAL-V15-V77-README

Purpose

This release folder contains the official v77 documentation set for Runtime Ecosystem Part 2.

Scope

- Covers Model Runtime.
- Covers Service Runtime.

Acceptance

- All documents contain implementation-level guidance.
- The release PDF and ZIP are generated and verified.
- MANIFEST and RELEASE_NOTES are updated for v77.

Runtime Ecosystem Part 2 Index

- [guide-15-77-01-model-runtime.md](#) - Model Runtime
- [guide-15-77-02-service-runtime.md](#) - Service Runtime

Model Runtime

Title: Model Runtime

Version: v77 draft

Status: Draft

Document ID: MAL-V15-MODEL-RUNTIME-01

Purpose

Provide official implementation guidance for Model Runtime inside Mariam AI Enterprise OS.

Scope

- Defines the recommended workflow, required evidence, roles, inputs, outputs, interfaces, and acceptance checks.
- Applies to architecture, engineering, operations, testing, governance, and release work connected to this topic.
- Excludes temporary experiments unless they are promoted into official documentation.

Responsibilities

- Owners maintain the document, examples, and acceptance evidence.
- Implementers follow the workflow and record deviations.
- Reviewers verify security, quality, traceability, and operational readiness.
- Operators confirm deployment, monitoring, recovery, and maintenance requirements where relevant.

Operating Model

- Plan the work from documented requirements and trace it to Blueprint or Specifications.
- Implement in small increments with tests, observability, and rollback behavior.
- Review changes against standards, threat model, failure modes, and acceptance evidence.
- Release only when documentation, tests, and operational notes are current.

Inputs

- Architecture Library documents, implementation tickets, configuration, schemas, policies, runtime context, and stakeholder constraints.
- Prior release notes, known risks, incident learnings, test results, and acceptance criteria.

Outputs

- Updated documentation, implementation plans, review evidence, test reports, operational runbooks, and release artifacts.
- Traceable decisions that explain why the selected approach is correct for Mariam.

Interfaces

- Git repository, CI/CD pipeline, runtime services, event bus, storage, dashboards, issue tracker, and release packaging.

- Human interfaces: review boards, chief office, department leads, operators, testers, and governance approvers.

Security

- Apply least privilege, secret isolation, signed releases where applicable, and audit-ready evidence.
- Do not expose tenant data, credentials, customer materials, private model artifacts, or privileged runtime outputs.
- Escalate security-sensitive deviations before implementation or release.

Metrics

- Lead time, review time, test pass rate, defect leakage, incident count, recovery time, documentation freshness, and acceptance completion.
- Topic-specific service metrics must be linked to dashboards and release notes.

Testing

- Verify expected behavior, failure behavior, recovery behavior, and evidence generation.
- Include automated tests where implementation exists and manual acceptance checks where documentation is the deliverable.
- Record test gaps explicitly in release notes.

Acceptance Criteria

- The workflow is clear enough for a new team member to execute without oral context.
- Required inputs, outputs, owners, interfaces, tests, and evidence are documented.
- Risks and security considerations are visible before release.
- The document is linked from INDEX, TRACEABILITY, ACCEPTANCE, MANIFEST, and release notes.

Traceability

Source	Relationship
MAL-V15-MODEL-RUNTIME-01	Governs Model Runtime.
Volume 05	Engineering standards reference.
RELEASE_NOTES_v77_draft.md	Release evidence.

Version History

Version	Date	Status	Notes
v77 draft	2026-06-29	Draft	Added official documentation for Model Runtime.

Service Runtime

Title: Service Runtime

Version: v77 draft

Status: Draft

Document ID: MAL-V15-SERVICE-RUNTIME-02

Purpose

Provide official implementation guidance for Service Runtime inside Mariam AI Enterprise OS.

Scope

- Defines the recommended workflow, required evidence, roles, inputs, outputs, interfaces, and acceptance checks.
- Applies to architecture, engineering, operations, testing, governance, and release work connected to this topic.
- Excludes temporary experiments unless they are promoted into official documentation.

Responsibilities

- Owners maintain the document, examples, and acceptance evidence.
- Implementers follow the workflow and record deviations.
- Reviewers verify security, quality, traceability, and operational readiness.
- Operators confirm deployment, monitoring, recovery, and maintenance requirements where relevant.

Operating Model

- Plan the work from documented requirements and trace it to Blueprint or Specifications.
- Implement in small increments with tests, observability, and rollback behavior.
- Review changes against standards, threat model, failure modes, and acceptance evidence.
- Release only when documentation, tests, and operational notes are current.

Inputs

- Architecture Library documents, implementation tickets, configuration, schemas, policies, runtime context, and stakeholder constraints.
- Prior release notes, known risks, incident learnings, test results, and acceptance criteria.

Outputs

- Updated documentation, implementation plans, review evidence, test reports, operational runbooks, and release artifacts.
- Traceable decisions that explain why the selected approach is correct for Mariam.

Interfaces

- Git repository, CI/CD pipeline, runtime services, event bus, storage, dashboards, issue tracker, and release packaging.

- Human interfaces: review boards, chief office, department leads, operators, testers, and governance approvers.

Security

- Apply least privilege, secret isolation, signed releases where applicable, and audit-ready evidence.
- Do not expose tenant data, credentials, customer materials, private model artifacts, or privileged runtime outputs.
- Escalate security-sensitive deviations before implementation or release.

Metrics

- Lead time, review time, test pass rate, defect leakage, incident count, recovery time, documentation freshness, and acceptance completion.
- Topic-specific service metrics must be linked to dashboards and release notes.

Testing

- Verify expected behavior, failure behavior, recovery behavior, and evidence generation.
- Include automated tests where implementation exists and manual acceptance checks where documentation is the deliverable.
- Record test gaps explicitly in release notes.

Acceptance Criteria

- The workflow is clear enough for a new team member to execute without oral context.
- Required inputs, outputs, owners, interfaces, tests, and evidence are documented.
- Risks and security considerations are visible before release.
- The document is linked from INDEX, TRACEABILITY, ACCEPTANCE, MANIFEST, and release notes.

Traceability

Source	Relationship
MAL-V15-SERVICE-RUNTIME-02	Governs Service Runtime.
Volume 05	Engineering standards reference.
RELEASE_NOTES_v77_draft.md	Release evidence.

Version History

Version	Date	Status	Notes
v77 draft	2026-06-29	Draft	Added official documentation for Service Runtime.